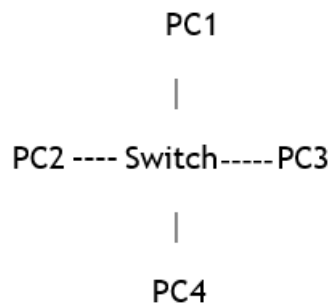


Network Topologies

Task 1.

Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

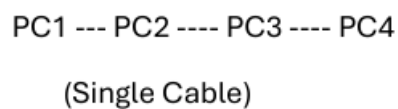
A) Star Topology



Example: WhatsApp

Reason: WhatsApp users connect to central servers that manage messages, making communication fast and easy to manage.

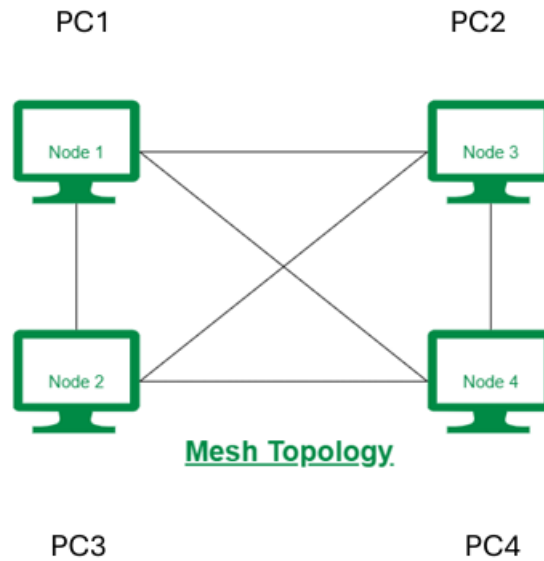
B) Bus Topology



Example: Small office file sharing

Reason: All computers share one communication cable, making it inexpensive for small networks.

C) Mesh Topology



Example: IPL Live Streaming (Content Delivery Network)

Reason: Multiple connections provide backup paths, ensuring continuous streaming even if one connection fails.

Task 2.

Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Topology	Pros	Cons	Use	App/Service Example
Star	1. Easy to install and manage 2. Easy to troubleshoot 3. Failure of one cable affects only one device 4. Easy to add new devices	1. Central switch failure affects the network 2. Requires more cable 3. More expensive than bus 4. Depends on a central device	Offices, schools, computer labs, warehouses	Zomato – office systems can use star-based LANs for easy management
Bus	1. Low installation cost 2. Uses less cable 3. Simple design 4. Suitable for small networks	1. Main cable failure affects the entire network 2. Difficult to troubleshoot 3. Performance decreases with more devices 4. Limited scalability	Small or older LANs	Older Ethernet networks – used a shared backbone cable
Ring	1. Data travels in an organized manner 2. No central switch required 3. Equal opportunity for devices to transmit 4. Predictable network performance	1. Failure of a link/device can disrupt communication 2. Difficult to troubleshoot 3. Adding/removing devices can be difficult 4. Less flexible than star	Legacy networks and specialized communication systems	Older Token Ring networks – used controlled access around a ring

Topology	Pros	Cons	Use	App/Service Example
Mesh	1. Very reliable 2. Multiple paths for data 3. Excellent fault tolerance 4. High availability	1. Expensive to install 2. Requires many connections 3. Complex configuration 4. Difficult maintenance	Banking, backbone, critical infrastructure	Paytm/Banking systems – redundant network paths help maintain availability

Task 3.

Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

Topology: Star Topology

Reason:

A star topology is the best choice because all players connect to central game servers. It provides fast communication, easy maintenance, and supports thousands of players simultaneously. If one player's connection fails, other players continue without interruption. Although server costs are higher, the reliability and performance make it ideal for games like PUBG and Free Fire.

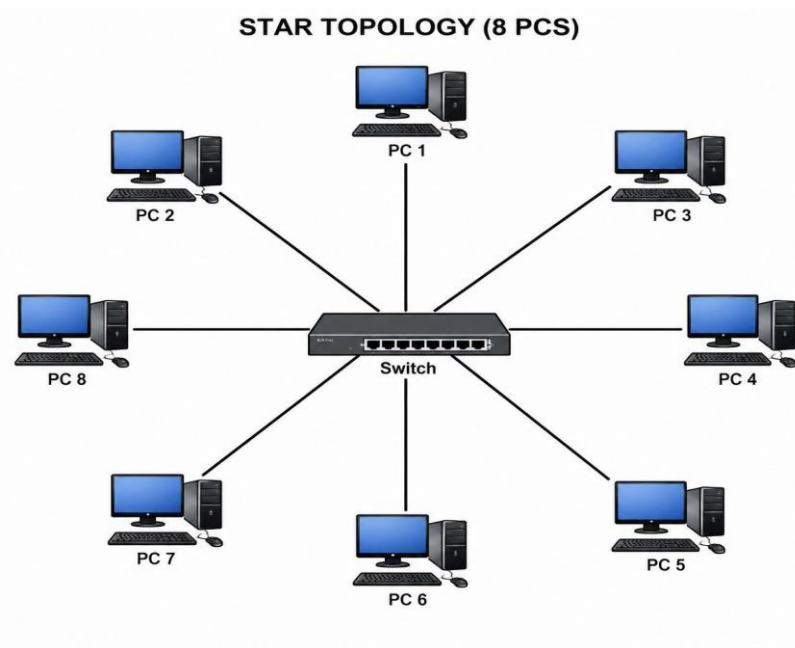
Task 4.

Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3

lines.

Hint: Consider factors like cost, ease of troubleshooting, and future expansion.

Topology: Star Topology



Reason

A star topology is the best choice because it is affordable, easy to troubleshoot, and simple to expand as the warehouse grows. If one computer or cable fails, the remaining computers continue working normally, making it reliable for order processing and inventory management.